

Measuring Ground Truth

Limb segment volume, and how to measure it so that
a disagreement with the pipeline means something

The short version

Measure circumferences with a tape and sum truncated cones.
Keep water displacement as a cross-check, and only with an
overflow spout.

What has to be true of a ground truth

The pipeline reports a volume. Nothing in it can tell you whether that volume is right. Only a physical measurement of the same limb can.

1. THE SAME QUANTITY THE PIPELINE REPORTS

The pipeline measures either the limb below one marker band or the segment between two of them. Those differ by more than a litre. A truth measured foot-to-band compared against a run cut band-to-band is not a 30% error — it is two different numbers.

2. REPEATABLE

A method whose run-to-run spread is larger than the error being investigated cannot resolve that error, however careful the arithmetic afterwards.

3. A BIAS YOU CAN NAME AND SIZE

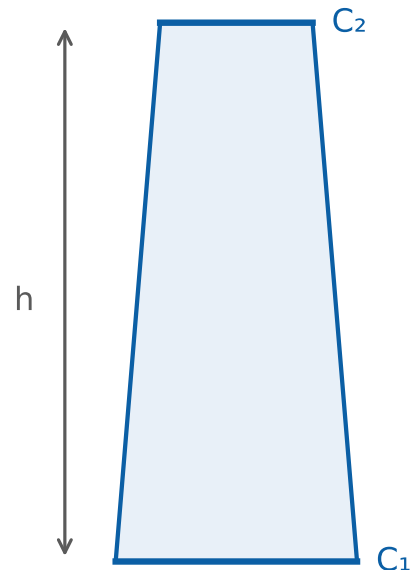
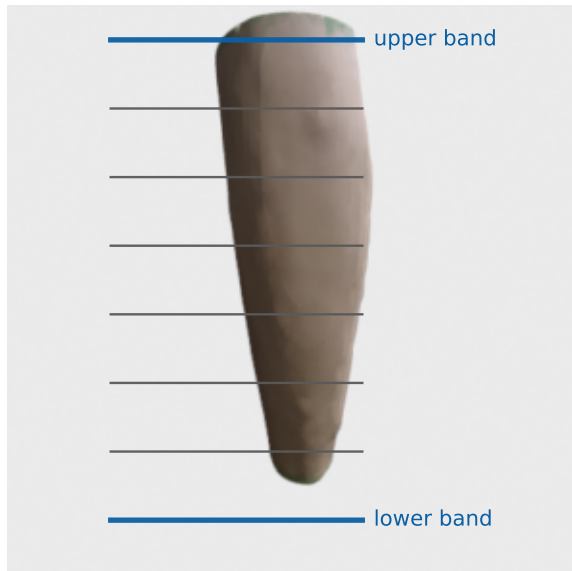
Every method is biased. A bias you can size is a correction. A bias you cannot is an unknown that contaminates every comparison drawn against it.

Method 1 — circumferences and the disc model

Preferred. Nothing is immersed, so none of the water errors apply.

each interval is a truncated cone

the segment between the bands (inputs/1_left)
marked every 4 cm



$$V_{slice} = \frac{h (C_1^2 + C_1 C_2 + C_2^2)}{12\pi}$$

PROCEDURE

1. Mark the limb every 4 cm from the lower band to the upper band.
Mark the skin, so the same heights can be found again.
2. Measure the circumference at each mark. Same tape, same tension,
flat against the skin without compressing it.
3. Record each circumference with its height above the lower band.
4. Repeat the whole set three times. Keep all three.

tools/limb_volume_from_tape.py sums the slices from a CSV.

WHAT IT GETS WRONG

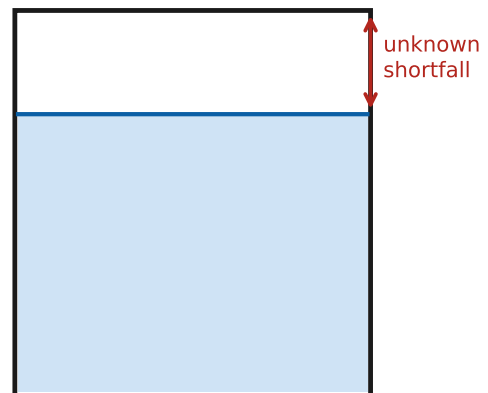
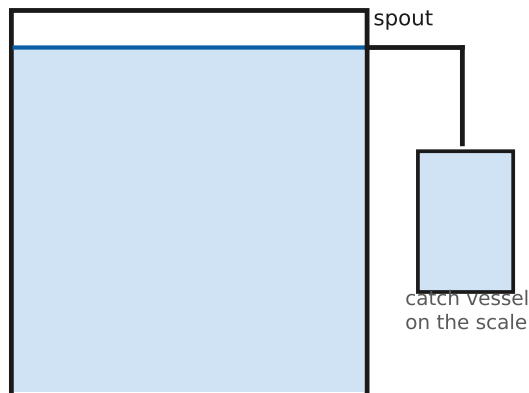
The model treats each cross-section as a circle. A calf is elliptical, and a circle of equal circumference encloses more area, so it reads high — about 2% at the 1.3 axis ratio measured at these ankles. That bias is constant, which is what makes it usable.

Method 2 — water displacement

Cross-check only. One gram of water is 1.00 mL to within 0.3% at room temperature.

RIGHT — spout, catch the overflow

WRONG — fill to the rim, re-weigh the tub



PROCEDURE

1. Use a container with an overflow spout at a fixed height.
2. Fill until the spout stops dripping on its own. This is the only way to put the starting level at a defined point, and it is the step that matters most.
3. Put the catch vessel on the scale and tare it.
4. Lower the limb until the water surface sits at the marker band. Hold still — muscle contraction changes limb volume.
5. Wait for the spout to stop dripping.
6. Weigh the caught water. Grams read as cm^3 .

For a segment, measure to each band separately and subtract.

WHY THE CATCH VESSEL, NOT THE TUB

Re-weighing the tub counts the film of water the limb carries out:

$$W_1 - W_2 = V_{\text{displaced}} + V_{\text{film}}$$

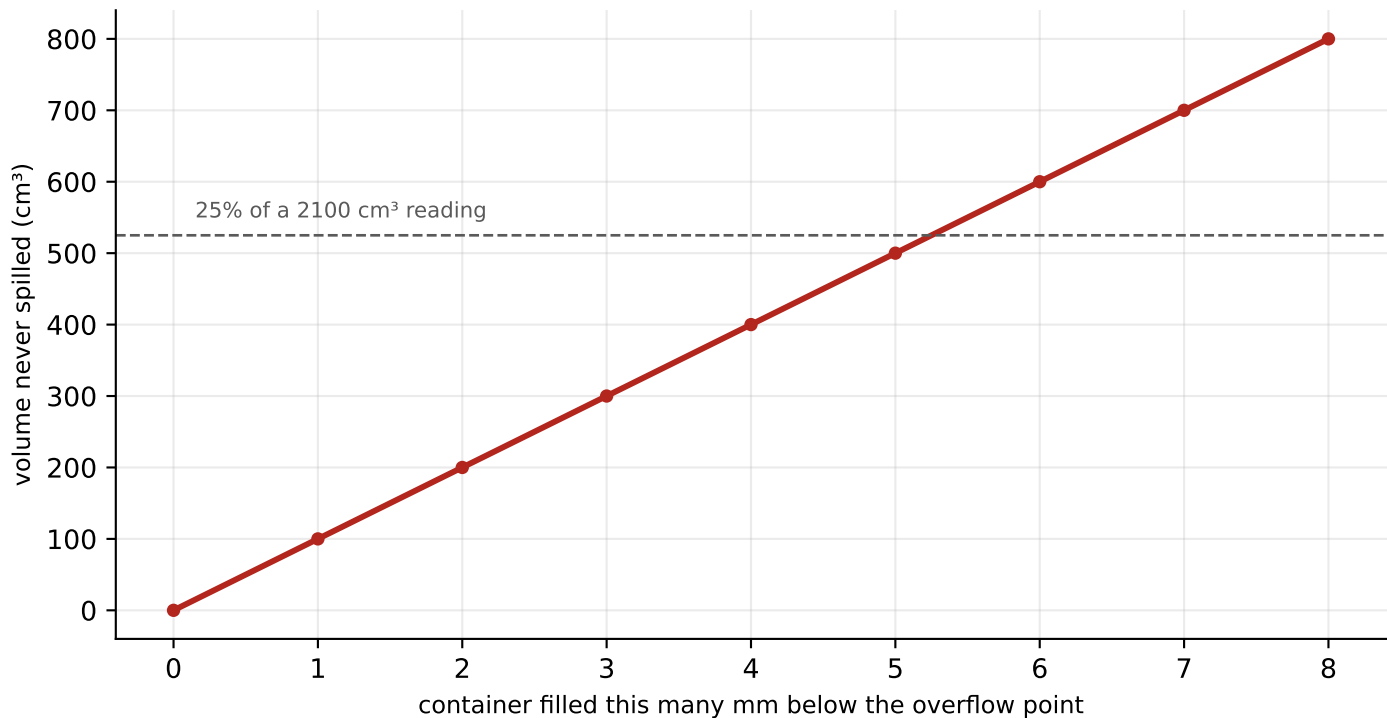
A wet calf carries 20–50 g. That is 1–2% of a 2100 cm^3 measurement to the upper band, but 7–17% of a 300 cm^3 measurement to the ankle — which is why the below-lower-band column is the least trustworthy in inputs/groundtruth0-6.csv.

Catching the overflow avoids it: the spill happens during immersion, the film leaves afterwards.

The errors, sized

Both were derived from this project's own measurements on 2026-09-04.

Fill-level shortfall — the dominant error without a spout



A container wide enough for a leg has roughly a 1000 cm² water surface, so every millimetre below the overflow point is 100 cm³ that never spills and is invisible to the scale. Without a spout, being 5 mm out is easy: surface tension lets water stand proud of a rim before it breaks, then it runs and settles somewhere below. The starting point is both unknown and different every time.

SCALE TILT — real, but too small to be the cause

A load cell reads the force component along its own axis, so a tilted scale scales both weighings, and their difference, by about $\cos \theta$:

tilt 5° → reading -0.4%
tilt 10° → reading -1.5%
tilt 20° → reading -6.0%
tilt 41° → reading -25%

Explaining a 25% discrepancy by tilt alone needs about 41°, which you would not miss. Level the scale and stop thinking about it.

TWO CHECKS WORTH RUNNING ONCE

Weigh a known volume. Pour a measured 2000 mL from a graduated flask into the catch vessel. If the scale reads 2000 g, the scale and its levelling are no longer suspects. Two minutes.

Repeat one limb three times without changing anything. Close agreement means a constant bias; scatter means the setup moves between

What to record

One row per limb, so the comparison scripts read it directly.

no	subject number	
sex	M or W	
<side>_low_delta	displaced volume to the LOWER band, cm ³	
<side>_up_delta	displaced volume to the UPPER band, cm ³	
<side>_vol	segment between bands = up_delta - low_delta	
band_<side>_lower	taped girth at the lower band, cm	
band_<side>_upper	taped girth at the upper band, cm	
band_<side>_separation	tape distance between the bands, cm	← new
<side>_tape_vol	disc-model volume between the bands	← new

Both new columns earn their place.

band_<side>_separation is the one measurement that lets plane placement be checked without trusting the reconstruction: the pipeline reports where it cut, and a tape says how far apart the bands actually are. Without it, a cut in the wrong place and a limb reconstructed the wrong size look identical in the volume.

<side>_tape_vol carries the disc-model result alongside the displacement one, so the two methods sit on the same limb rather than one silently replacing the other.

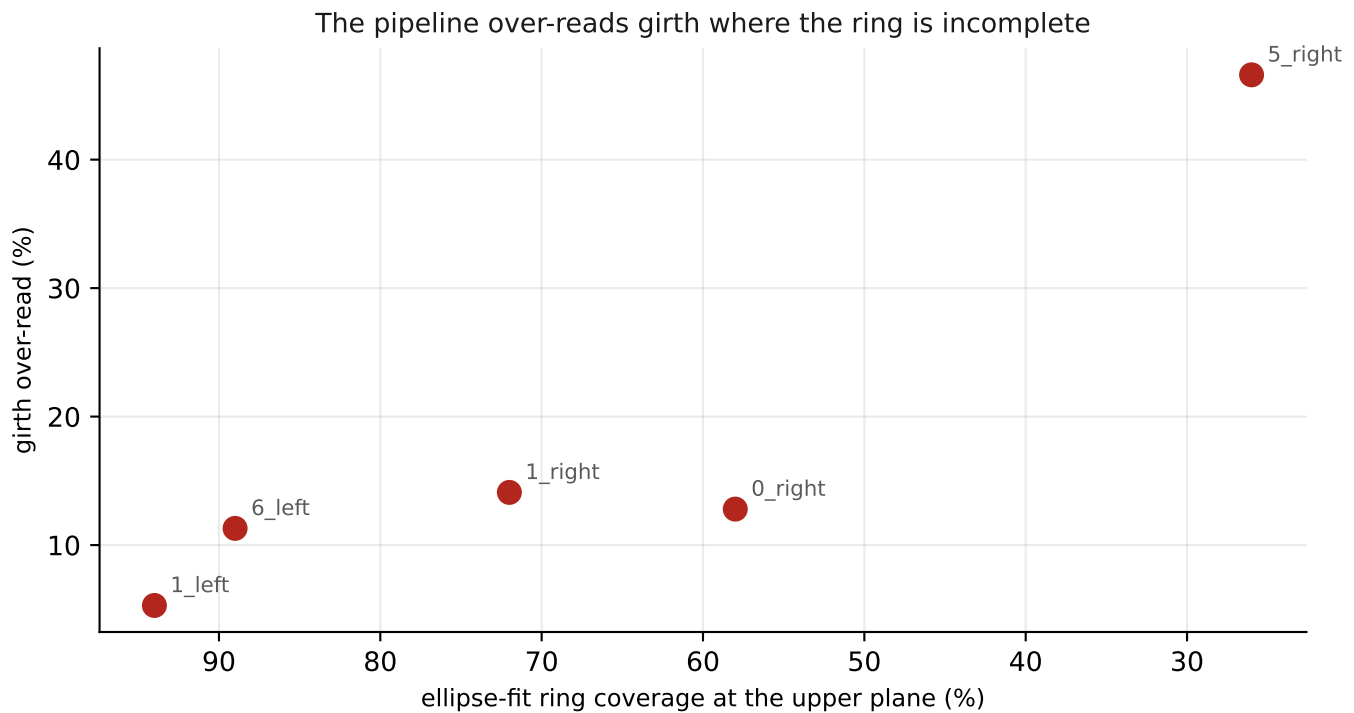
Record the full circumference profile as well — every height, every repeat, in a separate file:

```
subject,side,height_cm,circumference_cm
0,left,0.0,20.5
0,left,4.0,22.8
```

The summary row throws away exactly the part that localises error.

Open question, as of 2026-09-04

Two mechanisms, pointing opposite ways. Neither is isolated yet.



Fourteen captures were measured against inputs/groundtruth0-6.csv. The pipeline read +24.6% to +35.8% high on every capture that produced a sound two-plane cut. Two mechanisms have been identified:

THE PIPELINE OVER-READS GIRTH where the reconstruction ring is incomplete. Circumference is an ellipse fitted to a ring of surface points, and where the back of the calf did not reconstruct, the fit extrapolates across the gap. Volume goes as girth squared, so +11% girth is roughly +23% volume.

THE DISPLACEMENT METHOD UNDER-READS if the container is not at its overflow point, by 100 cm³ per millimetre of shortfall.

Neither has been isolated. Until one is, the difference between the two columns cannot be attributed to either side, and quoting it as a pipeline error overstates what is known.

The tape profile settles it. It shares no error mechanism with either — no water, no reconstruction — so it can adjudicate at every height rather than only at the end.